

Coding and Smart Systems Outreach Program

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Voice-Controlled LED

Hardware Components Required:

- Raspberry Pi 4 with GPIO extension
- Breadboard
- 1 LED
- 1 Resistor (220 Ω)
- 1 Black Wire
- 1 Yellow Wire
- **1 USB Microphone**

Step-by-step instructions to connect hardware components (**only perform steps highlighted in brown color**):

1. Plug the LED on the breadboard so that long and short legs are in different rows.
2. Plug the 220 Ω resistor with one end in the LED long-leg row and the other end in an empty row.
3. Use the yellow wire to connect that empty resistor-row to GPIO 17 (pin 11).
4. Use the black wire to connect the LED short leg row to GND pin 6.
- 5. Plug the USB microphone into any USB port on the Raspberry Pi.**
- 6. Power on the Pi, click on projects folder and then on Day1 folder.**
- 7. Click on the projects folder and then click on the voice_led.py file.**
- 8. Write the program code.**

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Python code (solution):

```
26 try:
27     #Tell program what is listening to your voice
28     with mic as source:
29         recognizer.adjust_for_ambient_noise(source)
30         print("Listening...")
31
32     #Loop to continue listening untill you speak
33     while True:
34         try:
35             audio = recognizer.listen(source, timeout=5, phrase_time_limit=3)
36             command = recognizer.recognize_google(audio).lower()
37
38             # What is missing
39             print(f"You said: {command}")
40
41             if "on" in command:
42                 #Two lines missing
43                 led.on()
44                 print("LED ON")
45             elif "off" in command:
46                 #Two lines missing
47                 led.off()
48                 print("LED OFF")
49             else:
50                 #One line missing
51                 print("Say 'on' or 'off'.")
52
53         except sr.WaitTimeoutError:
54             print("")
55
56         except sr.UnknownValueError:
57             print("")
58
59         except sr.RequestError:
60             print("Speech recognition service unavailable.")
61
62         sleep(0.5)
63
64 except KeyboardInterrupt:
65     pass
66
67 finally:
68     led.off()
69     print("Program terminated.")
```